

Aaron Broukhim

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Experience

Sep 2022 to June 2027	University of California San Diego - Graduate Researcher, PhD in CSE (Advisor: Nadir Weibel) <i>Thesis: preference-based reinforcement learning for generative speech</i> • Developed a LoRA fine-tuning method to condition a duplex model's audio output on internal uncertainty signals (semantic entropy, log-probability, hidden-state probes) without added inference latency. • Ran a between-subjects cross-modality preference study (n=94; 20 prompts each) showing that text and audio judgments of identical semantic content diverge; modeled rater bias with linear mixed-effects models and validated GPT-4o as a modality-dependent human proxy. • Collected a large-scale LLM-output preference dataset (n≈3,000; 18K+ pairwise judgments via Prolific/Qualtrics) for a study on content-moderation preferences across political affiliation. • Built a Mixture-of-Experts model with weight averaging, exploring compute/performance trade-offs.
June 2026 to Sept 2026	Adobe - Machine Learning Engineering Intern (Incoming) • Joining the Digital Experience AI & Product Engineering team to build ML/AI systems for marketing automation, including AI agents and reinforcement learning applications.
Jan 2022 to Aug 2022	Virufy - Associate Audio Machine Learning Engineer • Fine-tuned audio classifiers via transfer learning (TensorFlow, SK-Learn) for cough-based detection; ran feature/architecture search on SageMaker and peer-reviewed ML research.
July 2021 to Aug 2022	University of California San Diego - Undergraduate Student Researcher • Built logistic-regression and word-embedding models for hate-speech detection on Twitter; scraped data with Selenium; shipped a React/Node/Flask + AWS/SQL tool for non-experts to audit faulty ML systems.

Education

June 2027	University of California San Diego - Ph.D. in Computer Science & Engineering
June 2021	University of California San Diego - B.S. in Cognitive Science: Machine Learning

Selected Publications ([Google Scholar](#))

Accepted or Published	Kaufman et al., WARNING This Contains Misinformation: The Effect of Cognitive Factors, Beliefs, and Personality on Misinformation Warning Tag Attitudes, CSCW 2025 Kaufman et al., What Did My Car Say? Impact of Autonomous Vehicle Explanation Errors and Driving Context On Comfort, Reliance, Satisfaction, and Driving Confidence, CHI 2025
In Prep. or Under Review	<u>Broukhim</u> et al., Affect as Implicit Feedback in Speech, <i>in preparation</i> <u>Broukhim</u> et al., Aligning Perceived Confidence to Generated Speech, <i>in preparation</i> <u>Broukhim</u> et al., Social Emotion Dysregulation Evaluation Dataset and Models, <i>in preparation</i> Appel et al., Both Democrats and Republicans Prefer To Receive Multiple Perspectives in LLM Responses to Politically Divisive Topics, <i>under review</i> <u>Broukhim</u> et al., Same Words, Different Judgments: How Preferences Vary Across Modalities, <i>under review</i> <u>Broukhim</u> et al., Preference-Based Learning in Audio Applications: A Systematic Analysis, <i>under review</i>

Skills

Tech Skills	Languages: Python, SQL, Java, C++, C, JavaScript, HTML/CSS ML/Frameworks: PyTorch, TensorFlow, Keras, SK-Learn, HuggingFace, NumPy, Pandas, Kubernetes Infra/Tools: AWS, SageMaker, Git, Firebase, MySQL, React/Node/Flask, Selenium
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